

BELO HORIZONTE CONFINS, Brazil

WMO: 835870

Lat: 19.634S

Lon: 43.969W

Elev: 828

StdP: 91.77

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 10.8	(c) 11.8	(d) 3.1	(e) 5.2	(f) 23.3	(g) 5.0	(h) 6.0	(i) 22.0	(j) 8.2	(k) 19.6	(l) 7.5	(m) 19.5	(n) 2.0	(o) 180	(p) 0.322

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 9.7	(c) 32.1	(d) 19.6	(e) 31.0	(f) 19.9	(g) 30.1	(h) 19.9	(i) 23.6	(j) 27.2	(k) 22.9	(l) 26.9	(m) 22.3	(n) 26.5	(o) 3.4	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	19.4	25.1	22.0	18.4	24.4	21.2	17.5	23.7	75.1	27.4	72.4	27.0	70.0	26.4	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.6	(b) 6.7	(c) 6.0	(e) 8.0	(f) 34.3	(g) 1.4	(h) 1.5	(i) 7.0	(j) 35.3	(k) 6.1	(l) 36.2	(m) 5.3	(n) 37.0	(o) 4.3	(p) 38.1		
(a) 7.6	(b) 6.7	(c) 6.0	(e) 8.0	(f) 34.3	(g) 1.4	(h) 1.5	(i) 7.0	(j) 35.3	(k) 6.1	(l) 36.2	(m) 5.3	(n) 37.0	(o) 4.3	(p) 38.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.5	23.5	23.7	23.1	22.0	19.8	18.7	18.5	19.8	21.5	22.8	22.3	22.9
	DBStd	2.67	1.92	1.73	1.64	1.84	1.92	1.77	1.82	2.06	2.37	2.46	2.30	1.91
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	61	0	0	0	1	9	16	19	8	4	2	2	0
	CDD10.0	4209	417	383	404	359	304	262	264	304	344	398	368	400
	CDD18.3	1231	159	150	146	111	54	28	26	54	98	142	121	142
	CDH23.3	8008	1041	1032	847	611	283	172	194	441	774	1076	715	824
	CDH26.7	2028	302	301	191	108	24	7	13	85	230	363	190	213
Wind	WSAvg	2.9	2.8	2.8	2.7	2.7	2.5	2.5	2.7	3.1	3.3	3.4	3.1	2.9
Precipitation	PrecAvg	1344	292	163	152	47	27	7	7	12	44	76	223	305
	PrecMax	1650	640	320	352	116	84	40	42	46	92	136	383	588
	PrecMin	914	62	50	29	10	3	0	0	0	0	25	119	149
	PrecStd	192	163	74	86	34	18	11	11	15	28	34	66	104
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	32.2	31.1	30.1	28.9	27.5	28.1	30.9	33.1	33.9	32.2	31.8
		MCWB	21.0	21.8	21.5	20.3	19.5	17.6	16.7	17.3	18.0	18.3	19.7	20.6
	2%	DB	31.0	31.0	29.9	29.0	27.1	26.2	26.8	28.9	31.0	31.9	30.2	30.1
		MCWB	20.5	21.2	21.3	20.6	18.8	17.3	16.3	17.1	17.5	18.6	20.0	20.6
	5%	DB	29.8	29.9	28.9	28.0	26.1	25.2	25.2	27.2	29.2	30.2	28.9	29.0
		MCWB	20.7	20.9	21.2	20.4	18.4	17.0	16.3	16.3	17.5	18.8	20.1	20.8
	10%	DB	28.2	28.8	27.8	26.9	25.0	24.1	24.2	25.9	27.8	28.8	27.2	27.9
		MCWB	20.6	20.7	20.7	20.2	18.1	16.7	15.8	16.2	17.3	18.8	19.9	20.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.5	24.2	24.3	23.1	21.5	19.9	19.0	20.0	21.0	22.4	23.1	23.9
		MCDB	27.5	27.9	27.4	26.7	25.8	24.9	24.2	26.2	26.3	27.4	26.8	27.4
	2%	WB	23.4	23.3	23.1	22.1	20.6	18.8	18.2	18.8	20.1	21.1	22.2	22.9
		MCDB	26.8	27.6	26.9	26.4	25.0	23.6	23.8	24.8	25.8	26.3	26.1	26.6
	5%	WB	22.6	22.5	22.3	21.5	19.9	18.2	17.4	18.0	19.2	20.4	21.4	22.2
		MCDB	26.3	27.2	26.4	25.8	24.3	23.1	23.1	24.2	25.0	26.0	25.5	26.0
	10%	WB	21.8	21.7	21.6	20.8	19.0	17.4	16.8	17.2	18.6	19.9	20.8	21.5
		MCDB	25.7	26.4	25.7	24.9	22.9	22.4	22.4	23.4	24.5	25.7	24.7	25.2
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.7	9.2	9.7	10.4	11.0	11.6	12.1	11.6	10.7	9.0	8.8
		MCDBR	11.1	11.2	10.8	10.9	11.6	12.3	13.1	13.9	13.9	13.0	11.4	10.9
		MCWBR	3.9	3.7	3.9	4.0	4.5	4.7	4.9	4.5	4.2	4.0	3.7	3.8
	5% WB	MCDBR	9.1	10.0	9.6	9.9	10.2	11.1	11.5	12.1	11.1	10.6	9.3	9.2
MCWBR		4.6	4.5	4.4	4.3	4.7	4.9	4.8	4.5	4.3	4.1	4.1	4.4	
Clear-Sky Solar Irradiance	taub	0.387	0.377	0.372	0.352	0.334	0.323	0.327	0.357	0.439	0.436	0.399	0.391	
	taud	2.456	2.472	2.509	2.527	2.546	2.576	2.537	2.429	2.197	2.259	2.403	2.444	
	Ebn at Noon	957	956	936	917	898	891	896	898	859	890	940	954	
	Edh at Noon	121	117	109	100	91	85	90	108	146	143	127	122	
All-Sky Solar Radiation	RadAvg	5.88	6.01	5.33	4.91	4.28	4.03	4.30	5.03	5.38	5.65	5.20	5.49	
	RadStd	0.86	0.65	0.47	0.30	0.24	0.25	0.24	0.33	0.53	0.66	0.50	0.57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.84	-0.74	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	-0.84	-0.74	N/A	N/A	N/A	N/A

Nomenclature: See separate page